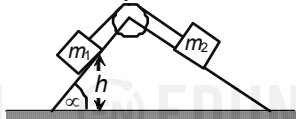
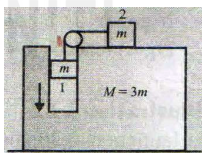


SUBJECTIVE TYPE

1. Two blocks of masses m_1 and m_2 are connected by a massless, inextensible, cord that is passed over a massless pulley A . The blocks slide along the smooth sides of a right angled wedge of mass m , which rests on a smooth horizontal plane. Find the distance covered by the wedge on the horizontal plane till the mass m_1 is lowered by the vertical distance h .

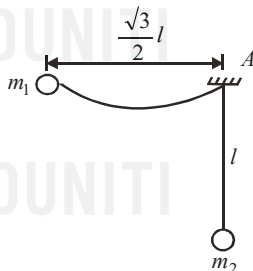


2. A wedge of mass $M = 3m$ and having a vertical slot in it is placed on a horizontal surface. Two blocks each of mass m are arranged as shown in the figure. The system is released from rest. All surface are smooth. Calculate the speed of the wedge when block 1 comes down a distance h .



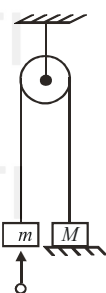
3. Two balls of mass $m_1 = 100 \text{ g}$ and $m_2 = 300 \text{ g}$ are suspended from point A by two equal inextensible threads, each of length $\ell = 32/35 \text{ m}$.

Ball of mass m_1 is drawn aside and held at the same level as A but at a distance $\sqrt{3}\ell/2$ from A , as shown in fig. When ball m_1 is released, it collides elastically with the stationary ball of mass m_2 . Calculate



- velocity u_1 with which the ball of mass m_1 collides with the other ball
 - maximum rise of the ball of mass m_2
- (Take $g = 10 \text{ ms}^{-2}$)

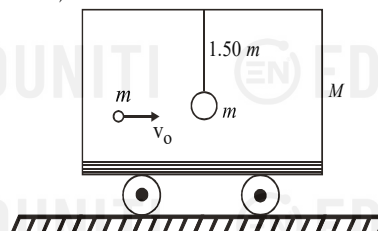
4. A light flexible thread passes over a small, frictionless pulley. Two blocks of mass $m = 1 \text{ kg}$ and $M = 3 \text{ kg}$ are attached with the thread as shown in fig. Heavier block rests on a slab. A bullet of mass $m_0 = 1 \text{ kg}$, moving upwards with velocity $v_0 = 10 \text{ ms}^{-1}$, collides with the hanging block at time $t = 0$. Calculate the maximum height ascended by M when it is jerked into motion



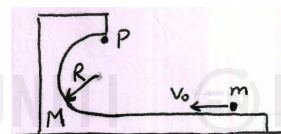
- if the bullet gets stuck in the hanging block
- if the bullet collides with the hanging block elastically

5. A ball of mass $m = 1 \text{ kg}$ is hung vertically by thread of length $l = 1.50 \text{ m}$. Upper end of the thread is attached to the ceiling of a trolley of mass $M = 4 \text{ kg}$. Initially, trolley is stationary and is free to move along horizontal rails without friction. A bullet of mass $m = 1 \text{ kg}$, moving horizontally with velocity $v_0 = 6 \text{ ms}^{-1}$, collides with the ball and gets stuck with it. As a result, thread starts to deflect towards right. Calculate its maximum deflection with the vertical.

(Take $g = 10 \text{ ms}^{-2}$)



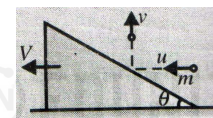
6. A small ball of mass m is projected with a minimum horizontal velocity v_0 on a block of mass M so that it will reach the highest point P of the block. If all surfaces are smooth, find the value of v_0 .



7. A small steel ball A is suspended by an inextensible thread of length $\ell = 1.5 \text{ m}$ from O . Another identical ball is thrown vertically downwards such that its surface remains just in contact with thread during downward motion and collides elastically with the suspended ball. If the suspended ball just completes vertical circle after collision, calculate the velocity of the falling ball just before collision.



8. A ball of mass ' m ' moving horizontally with velocity ' u ' hits a wedge of mass ' M '. The wedge is situated on a smooth horizontal surface. After striking with wedge, the ball starts moving in vertical direction and the wedge starts moving in horizontal plane. Calculate

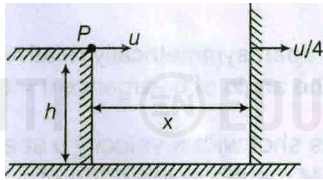


- the velocity V of wedge.
- the velocity v at which the ball moves in vertical direction.
- the impulse imparted by the ball to the wedge.
- the coefficient of restitution e ?

PARAGRAPH TYPE

Paragraph for Question Nos. 9 to 10

A particle is thrown with a velocity u from a height h horizontally towards a vertical wall which is moving away with a speed $u/4$ as shown in figure. The particle returns to the point of projection after suffering two elastic collisions (one with the wall and the other with the ground).

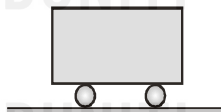


9. The total time of flight is
- (a) $2\sqrt{2h/g}$ (b) $2\sqrt{h/g}$
 (c) $\sqrt{2h/g}$ (d) $4\sqrt{2h/g}$
10. What is the initial separation x between the particle and the wall?
- (a) $2u\sqrt{h/2g}$ (b) $u\sqrt{h/2g}$
 (c) $u\sqrt{2h/g}$ (d) $u\sqrt{h/g}$

Paragraph for Question Nos. 11 to 13

When the mass of a system is variable, a thrust force acts on it in addition to all other forces acting on it. This thrust force is given by $\vec{F} = \vec{v}_r (dm/dt)$ where m is the instantaneous mass of the system and $\vec{v}_r$ is the relative velocity with which the mass dm either enters or leaves the system. A car has total mass 50 kg. Gases are ejected from this backwards with relative velocity 20 m/s.

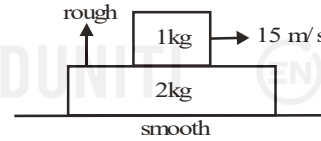
The rate of ejection of gas is 2 kg/s. Total mass of gas is 20 kg. Coefficient of friction between the car and road is $\mu = 0.1$ (Take $g = 10 \text{ m/s}^2$)



11. Car will start moving after time
- (a) 4 s (b) 10 s
 (c) 5 s (d) 8 s
12. Maximum speed of car will be (Take $\ln(4/3) = 0.28$)
- (a) 0.6 m/s (b) 0.8 m/s
 (c) 1.0 m/s (d) 1.2 m/s
13. Car will stop after (from starting)
- (a) 12.2 s (b) 6.4 s
 (c) 10.6 s (d) 5.8 s

Paragraph for Question Nos. 14 to 16

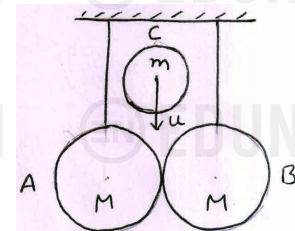
A 1 kg block is given a velocity of 15 m/s towards right over a very long rough plank of mass 2 kg as shown in figure.



14. The correct graph showing linear momentum of 1 kg (i.e. p_1) and of 2 kg (i.e. p_2) versus time is
- (a) (b)
 (c) (d)
15. If coefficient of friction between the two blocks is equal to 0.4, then magnitude of initial slope of p_1 versus t and p_2 versus t (in SI unit) will be (Take $g = 10 \text{ m/s}^2$)
- (a) 4 and 2 (b) 2 and 4
 (c) 4 and 4 (d) 2 and 2
16. Momentum of both the blocks is equal at time
- (a) 1.75 s (b) 1.875 s
 (c) 2.5 s (d) 1.25 s

Paragraph for Question Nos. 17 to 19

Two identical balls A and B, each of mass $M = 2 \text{ kg}$ and radius R , are suspended vertically from inextensible strings as shown in fig. The third ball C of mass $m = 1 \text{ kg}$ and radius $r = (\sqrt{2} - 1)R$ falls and hits A and B symmetrically with speed $u = 10 \text{ m/s}$. Speed of both A and B just after the collision is 3 m/s.



17. Impulse provided by each string during collision is
- (a) $6\sqrt{2} \text{ N s}$ (b) 12 N s
 (c) $3\sqrt{2} \text{ N s}$ (d) 6 N s
18. Speed of C just after collision is
- (a) 2 m/s (b) $2\sqrt{2} \text{ m/s}$
 (c) 5 m/s (d) $(\sqrt{2} - 1) \text{ m/s}$

19. The value of coefficient of restitution is

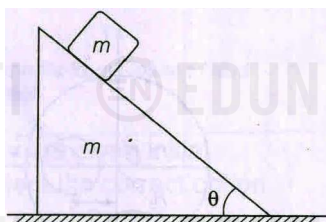
- (a) $1/4$ (b) $1/\sqrt{2}$
(c) $\sqrt{2} - 1$ (d) $1/2$

MORE THAN ONE CORRECT

20. A particle moving with kinetic energy = 3J makes an elastic head on collision with a stationary particle which has twice its mass. During the impact,

- (a) the minimum kinetic energy of the system is 1J.
(b) the maximum elastic potential energy of the system is 2J.
(c) momentum and total energy are conserved at every instant.
(d) the ratio of kinetic energy to potential energy of the system first decreases and then increases.

21. A block of mass m slides down on an inclined plane of a wedge of mass m (see figure). Friction is absent everywhere. Which of the following statement(s) is/are correct?



(a) Vertical component of acceleration of block is

$$\frac{g \cos^2 \theta}{1 + \sin^2 \theta} \text{ downwards}$$

(b) Acceleration of centre of mass of (block + wedge) is

$$\frac{g \sin^2 \theta}{1 + \sin^2 \theta} \text{ downwards}$$

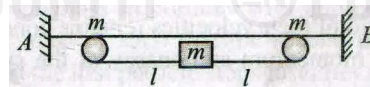
(c) Acceleration of centre of mass of (block + wedge) is

$$\frac{g \cos \theta}{1 + \sin^2 \theta} \text{ downwards}$$

(d) Vertical component of acceleration of block is

$$\frac{2g \sin^2 \theta}{1 + \sin^2 \theta} \text{ downwards}$$

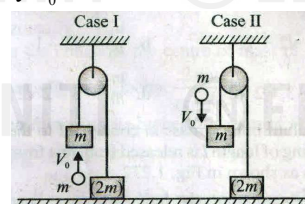
22. Two small rings, each of mass ' m ', are connected to the block of same mass ' m ' through an inextensible massless strings of length ' l ' each. Rings are constrained to move over smooth rod AB. Initially, the system is held at rest as shown in fig. Let u and v be the velocities of ring and block, respectively when string makes an angle 60° with the vertical. Then,



(a) $u = \sqrt{\frac{g\ell}{5}}$ (b) $u = \sqrt{\frac{8g\ell}{5}}$

(c) $v = \sqrt{3g\ell}$ (d) $v = \sqrt{\frac{3g\ell}{5}}$

23. Two masses $2m$ and m are connected by an inextensible light string. The string is passing over a light frictionless pulley. The mass $2m$ is resting on a surface and mass m is hanging in air as shown in fig. A particle of mass m strikes mass m from below in case (I) and from top in case (II) with a velocity v_0 and sticks to it.



- (a) The conservation of linear momentum can be applied in both the cases just before and just after collision.
(b) The conservation of linear momentum can be applied in case I but cannot be applied in case II just before and just after collision.
(c) The ratio of velocities of mass m just after collision in first and second case is $1/2$.
(d) The ratio of velocities of mass m just after collision in first and second case is 2 .

24. Object A strikes the stationary object B with a certain given speed u head-on in an elastic collision. The mass of A is fixed, you may choose the mass of B appropriately for following cases. Then after the collision,

- (a) for B to have the greatest speed, choose $m_B = m_A$
(b) for B to have the greatest momentum, choose $m_B \gg m_A$
(c) for B to have the greatest speed, choose $m_B \ll m_A$
(d) for the maximum fraction of kinetic energy transfer, choose $m_B = m_A$



25. A small particle of mass m starts sliding down from rest along the smooth surface of a fixed hollow hemisphere of mass $M (= 4m)$. The distance of centre of mass of (particle + hemisphere) from centre O of hemisphere, when the particle separates from the surface

of hemisphere is $\frac{\sqrt{69}}{5\alpha}R$.
Find the value of α

